ARM PrimeCell

Static Memory Controller (PL092)

Release Note

PL092 RELN 0000 D

Open Access

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Feedback on the ARM PrimeCell Static Memory Controller

If you have any comments or suggestions about this product, please contact your supplier giving:

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Change history

Issue	Date	Change
A	05 June. 2001	First release.
B	08 Dec.2002	Second release for new version r1p2
C	09 June 2003	Release for version r1p3-00ltd0
D	04 September 2003	Release for version r1p3-01ltd0

Contents

1 PRODUCT DELIVERABLES

1.1 ARM Part Numbers for this product

Table 1.1-1 lists the ARM part numbers for the individual deliverables that form part of the delivery for this ARM PrimeCell peripheral:

Product deliverables for the ARM PrimeCell PL092 peripheral	Document number	Product code
Synthesizable VHDL RTL	N/A	PL092-MN-23001-r1p3-01ltd0
Synthesizable Verilog RTL	N/A	PL092-MN-22001-r1p3-01ltd0
Synthesis Scripts (Synopsys Design Compiler, Avant! CB25 cell library)	N/A	PL092-MN-01002-r1p3-01ltd0
Testbench - Functional Verification, VHDL Testbench - Integration test, VHDL	N/A N/A	PL092-MN-23010-r1p3-01ltd0 PL092-MN-23012-r1p3-01ltd0
Testbench - Functional Verification, Verilog	N/A	PL092-MN-22010-r1p3-01ltd0
Testbench - Integration test, Verilog	N/A	PL092-MN-22012-r1p3-01ltd0
Test vectors-BusTalk, functional verification	N/A	PL092-VE-01010-r1p3-01ltd0
Test vectors-BusTalk, integration test	N/A	PL092-VE-01012-r1p3-01ltd0
ARM PrimeCell Static Memory Controller (PL092) Technical Reference Manual (PDF)	ARM DDI 0203	PL092-DA-03001-r1p3-01ltd0
ARM PrimeCell Static Memory Controller (PL092) Design Manual (PDF)	PL092 DDES 0000	PL092-DC-02002-r1p3-01ltd0
ARM PrimeCell Static Memory Controller (PL092) Integration Manual (PDF)	PL092 INTM 0000	PL092-DC-10002-r1p3-01ltd0
ARM PrimeCell Static Memory Controller (PL092) Release Note (PDF)	PL092 RELN 0000	PL092-DC-06002-r1p3-01ltd0
ARM PrimeCell Static Memory Controller (PL092) Errata Notice (PDF)	PL092 DEI 0000	PL092-DC-11001-r1p3-01ltd0
ARM PrimeCell Static Memory Controller (PL092) Specific Userguide	PL092 USGD 0000	PL092-DC-08001-r1p3-01ltd0
The parts with numbers starting PL000 are delivered as a separate bundle PL000-BU-00004		
System Common and System Indirection Source Code	N/A	PL000-SW-02001-r9p0-00rel0
System Test Source Code	N/A	PL000-SW-02003-r9p0-00rel0
System Common Files Source Code	N/A	PL000-SW-02004-r9p0-00rel0
System Build Project Files	N/A	PL000-SW-02005-r9p0-00rel0
Interrupt Controller Object Code	N/A	PL001-SW-00000-r9p0-00rel0
Timer Object Code	N/A	PL002-SW-02001-r9p0-00rel0
Software Code Module	N/A	PL092-SW-02001-r1p3-01ltd0

Software Test Report	PL092_TREP_1000	PL092-DC-01001-r1p3-01ltd0
Software API Document	PL092_ISPC_1000	PL092-DC-02001-r1p3-01ltd0
Timer API Documents	PL002_ISPC_1000	PL002-DC-02001-r9p0-00rel0
System Structure and Integration Guide	PL000_DII_1000	PL000-DC-02000-r9p0-00rel0
System Common API Documents	PL000_ISPC_1001	PL000-DC-02001-r9p0-00rel0
System Indirection API Documents	PL000_ISPC_1002	PL000-DC-02001-r9p0-00rel0
System Test API Documents	PL000_ISPC_1003	PL000-DC-02003-r9p0-00rel0
System Common Files API Documents	PL000_ISPC_1004	PL000-DC-02004-r9p0-00rel0
Interrupt Controller API Documents	PL001_ISPC_1000	PL001-DC-02001-r9p0-00rel0

Table 1.1-1: ARM part numbers for ARM PrimeCell SMC PL092

Note Only the required deliverables will be supplied, as per the contractual agreement.

1.2 Reference Cell Library

A reference cell library (Avant! CB25 0.25µm standard cells) is supplied with this peripheral.

This library is supplied as a separate bundle, part PL000-BU-00003-2.03.

PL000-BU-00003-2.03/Avanti/CB25_1999.1.tar.Z
PL000-BU-00003-2.03/Avanti/CB18_1999.1.lst

1.3 Common Driver Bundle

A bundle containing software drivers, common to all PrimeCells is also supplied.

This bundle is part PL000-BU-00004 and is supplied as a single tarred and zipped file,

PL000-BU-00004-r9p0-00rel0.tgz

2 INSTALLATION

2.1 Introduction

Intellectual Property (IP) deliverables are collectively delivered as a single UNIX zipped tar file which has been security encrypted.

An ARM PrimeCell peripheral may be one of several products delivered as part of this single encrypted file. Each ARM PrimeCell peripheral comprises a set of zipped tar files which correspond to a unique part number quoted in the contractual agreement document. A corresponding list file details the files in each tar file

These installation instructions cover the following platform/operating system:

- UNIX

2.2 Disk space required

This installation will require 100 Mbytes of free disk space.

2.3 Installation procedure

You will have a single tarred, zipped and encrypted file of following format:

E.g.

Company Name-PrimeCell-PL092–Bundled-arm-download-WWWW.tgz.des

The installation procedure is summarised below:

2.3.1 Unpacking the shipment

The following steps describe how to unpack the separate products delivered in this shipment.

The first three steps may have already been done, ie the shipment from ARM may have already been decrypted and unzipped. If this is the case then go to step 4, else start with step 1.

1. **Relocate the shipment file**

   Relocate the encrypted file (for example <file>.tgz.des, if DES encryption has been used) to an appropriate location that has sufficient free disk space if PGP encrypted will need the relevant keys.

2. **Decrypt file**

   Follow the accompanying instructions for decryption of the file supplied, to generate a decrypted, compressed, tar file <file>.tgz

3. **Unzip file**

   Execute the UNIX unzip command to obtain the tar file <file>.tar:

   gunzip <file>.tgz

   where <file> is the name of the file supplied.

4. **Extract tar files**

   To extract the PrimeCell, execute the UNIX tar extract command:

   ```
   tar -xvf <file>.tar
   ```

This will generate the following three files

ARM_MANIFEST_XXXX.TXT – which lists the PrimeCell files delivered

ARM_DELIVERY_XXXX.TXT – which lists the ARM parts delivered

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And will unpack the PrimeCell into the top-level peripheral directory (e.g. smc_pl092)

The peripheral development environment is stored in two directory structures, a peripheral specific directory structure and a project global directory structure. The top level directory structures are shown in ***Figure 2-1.***

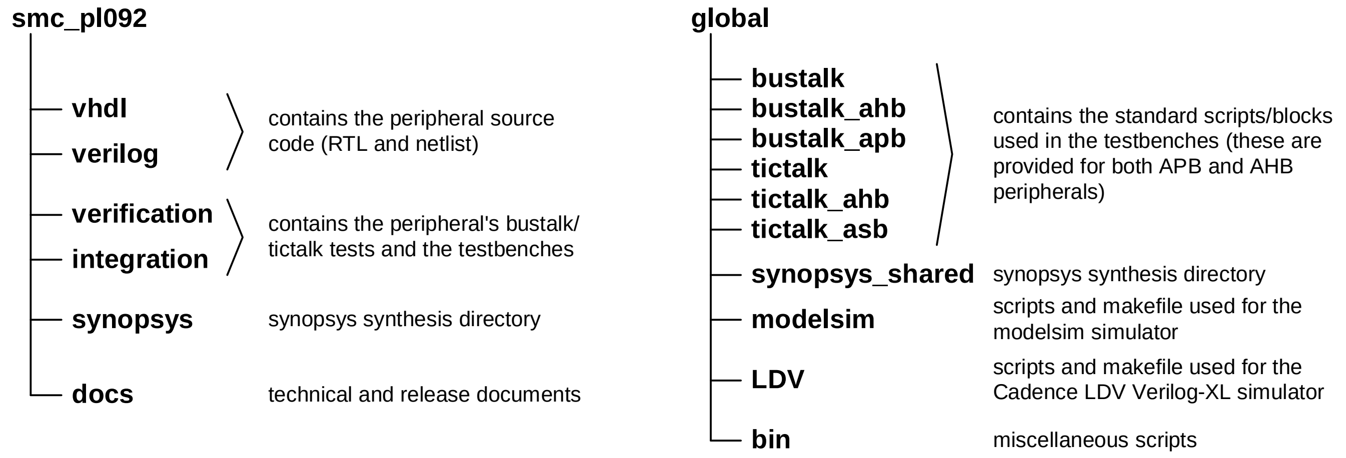

Figure 2-1. Peripheral Development Environment Directory Structure

2.3.2 Installation of the reference cell library

The reference cell library should be copied manually and tar extracted in the directory containing the PrimeCell peripherals. An example is shown below to copy the Avant! CB25 1999.1 cell library to the directory /ARM/PrimeCell_peripherals

```
cd /ARM/PrimeCell_peripherals
cp /PL000-BU-00003-2.03/Avanti/CB25_1999.1.tar.Z .
```

Execute the UNIX unzip and tar extract commands as below:

```
unzip CB25_1999.1.tar.Z
tar -xvf CB25_1999.1.tar
```

2.3.3 Installation of the Common Driver Bundle

A common software bundle PL000-BU-00004-r9p0-00rel0 is supplied with this delivery.

The files in the common driver bundle unzip and untar into a single directory PrimeCell_gen_sw.

The top level directory structures are shown in ***Figure 2-2.***

```
PrimeCell_gen_sw/
 ├── PL000_DII_1000.pdf
 ├── Documentation/
 └── Software/
      ├── apcommon/
      ├── apint/
      ├── apos/
      ├── aptest/
      ├── aptimer/
      └── build/
```

Figure 2-2 Common PrimeCell Driver Directory structure

3 DOCUMENTATION

3.1 Adobe PDF

Documents are supplied as “Adobe PDF” (Portable Document Format) files. These files are readable on most common computer platforms and operating systems using the appropriate file reader. A suitable file reader can be downloaded from the Adobe World Wide Web site at http://www.adobe.com. Select “Acrobat Reader" and download the reader for your computer platform/operating system.

3.2 Peripheral Documents Supplied

The following documents are supplied for the ARM PrimeCell Static Memory Controller PL092:
The documents files are located in the smc_pl092/docs/ directory.

ARM PrimeCell Static Memory Controller PL092
Technical Reference Manual ARM DDI 0203

The technical reference manual provides details for programming and interfacing to this ARM PrimeCell peripheral block.

ARM PrimeCell Static Memory Controller PL092
Integration Manual PL092 INTM 0000

The integration manual provides information required to integrate this ARM PrimeCell block within a typical industry standard ASIC design flow.

ARM PrimeCell Static Memory Controller PL092
Design Manual PL092 DDES 0000

The design manual describes the implementation details of this ARM PrimeCell block and the testbenches and test programs used for functional verification.

ARM PrimeCell Static Memory Controller PL092
Errata Notice PL092 DEI 0000

The errata notice describes details of any known issues with this ARM PrimeCell block and any workarounds for these issues. In most cases this document will be blank or may not be present.

ARM PrimeCell Static Memory Controller PL092
User Guide PL092 USGD 0000

The user guide describes the instructions on performing simulations and synthesis for this ARM PrimeCell block.

4 TRIAL TESTING

4.1 Product Maturity

This product is newly developed IP and approved for final release.
At the time of release, it has not been implemented in production silicon.

4.2 Technical data summary

The following is a summary of the key technical points from testing of the ARM PrimeCell Static Memory Controller PL092.

	VHDL RTL	Verilog RTL
Cell library	Avant! Passport 1999.3 CB25 v2.1 (0.25 µm)	Avant! Passport 1999.3 CB25 v2.1 (0.25 µm)
Total cell area, without scan, NAND-2 equivalents	9211	9279
Net interconnect area, without scan, NAND-2 equivalents	4803	4844
Total cell area, scan inserted,NAND-2 equivalents	11093	11145
Net interconnect area, scan inserted,NAND-2 equivalents	5995	6005
Clock frequency used	100 MHz	100MHz
Scan insertion figure % (number of 'D' flip-flops replaced)	100% (687)	100% (687)
Estimated scan fault coverage, collapsed.	99.81%	99.83%
Synthesis and scan test tools and version	Synopsys Design Compiler and Test Compiler 2001-08-SP2-2	Synopsys Design Compiler and Test Compiler 2001-08-SP2-2
Simulation tools and versions	Model Technology Inc, ModelSim v5.5d	Model Technology Inc, ModelSim v5.5d Cadence Verilog-XL 3.3ISR
Denali Memory models version	Denali 3.000-0044	Denali 3.000-0044
Global Peripheral Development Environment Version	Release REL1v7	

4.3 BusTalk Functional block verification test scenarios

Functional block verification has been performed with the following scenarios using the BusTalk testbench with free-running clocks.

4.3.1 Synchronous clocks

SMCCLK/PCLK	Simulations
100MHz (10ns period)	VHDL and Verilog, RTL and gate-level min/max SDF

For further details of functional verification testing, refer to the ARM PrimeCell Static Memory Controller (PL092) Design Manual. PL092-DDES-0000.pdf

5 DIFFERENCES FROM PREVIOUS VERSIONS

5.1 Differences between versions 1v1 and r1p2-00ltd0

The following issues have been fixed in release r1p2-00ltd0. These issues are described in the Errata Notice PL092-DEI-0000.pdf

Errata no. 2.1 HREADY Signal Driven HIGH Prematurely in AddrToggleTests.c

Errata no. 2.2 Deadlock Occurs When There is a Change in Memory Configuration

Errata no. 2.3 SMC Generates Incorrect Accesses When Using Different Sized Banks

Errata no. 2.4 A Write Following a Burst Transfer is Ignored When No Wait States Are Used

Errata no. 2.8 Error in nSMDATAEN generation logic

Errata no. 2.9 Error in IDLE cycle generation logic

Errata no. 2.10 Error occurs in AHB burst writes where HSIZE is less than memory width

Errata no. 2.11 HREADYOUT generated prematurely

Errata no. 2.12 Register write occurs with zero AHB wait states

Errata no. 2.13 A memory transfer takes place during an IDLE cycle with Burst Mode devices

Errata no. 2.14 nSMCBLSOUT and nSMCS may be incorrectly generated

Errata no. 2.15 SMCDATAOUT is generated one cycle late for memory writes with AHB size (HSIZE) greater than the memory size (MSIZE)

Errata no. 2.16 Issues with AHB WRAP4 transfers and end of burst internal signals

Errata no. 2.18 Transfer state machine signal not deasserted at end of Burst Mode Read

Errata no. 2.19 Burst Mode indicator incorrectly deasserted at the end of the first read of two read transfers

The following enhancement, documented as Errata no. 2.17 has been made to the r1p2-00ltd0 release.

The r1p2 release contains a hardware fix to ensure that the register values are valid, by loading the WST1/2 value into the read/write enable counters respectively if the WSTOEN/WEN values are programmed greater than the WST1/2 values. This is a feature enhancement to match other current PrimeCell memory controllers and is not a bug fix.

5.2 Differences between versions r1p2-00ltd0 and r1p3-00ltd0

The following issues have been fixed in this release. These issues are described in the Errata Notice PL092-DEI-0000.pdf

Errata no. 3.1 Booting is not possible from 16/32-bit devices with Byte Lane inputs on chip select 7

Errata no. 3.2 Issues with SMWAIT signal

Errata no. 3.3 Issues with DBI/EBI

5.3 Differences between versions r1p3-00ltd0 and r1p3-01ltd0

The following issue has been fixed in this release. This issue is described in the Errata Notice PL092-DEI-0000.pdf

Errata no. 4.1 SmcState Machine Locks in ST_BUFWR state and EbiCorner Case Modified

6 KNOWN ISSUES

At the time of release there are no known issues with PL092-r1p3-01ltd0

Any issues found subsequent to release will be documented in updates to the Errata Notice PL092 DEI 0000.

7 REVISION HISTORY

Date	Revision	Description
05 June 2001	REL 1v1	Initial release
1 Feb 2003	r1p2-00ltd0	Second release with bug-fixes
13 June 2003	r1p3-00ltd0	Third release.
04 September 2003	r1p3-01ltd0	Fourth Release with Ebi bug fix